Cloud-Cleared Radiances From Collocated Observations of Hyperspectral IR Sounder and Advanced Imager Onboard the Same Geostationary Platform

Xinya Gong, Zhenglong Li, Jun Li, Ruoying Yin, Wei Han, Lin Chen, and Di Di

***Abstract*— The Geostationary Interferometric Infrared Sounder (GIIRS) onboard China's Fengyun-4A (FY-4A) geostationary (GEO) meteorological satellite provides high-spectral-resolution infrared (IR) observations for targeted observing areas with high temporal resolution. Due to the high uncertainties in radiative transfer modeling of cloudy radiances, it is challenging to take full advantage of the thermodynamic information from GIIRS in all-sky conditions. The Advanced Geostationary Radiation Imager (AGRI) onboard the same platform provides a variety of cloud products with high spatial resolution. A bias-corrected optimal cloud-clearing (BCOCC) approach is introduced to generate GIIRS cloud-cleared radiances (CCRs) with the help of AGRI-collocated clear radiances (CLRs). The bias correction (BC) scheme is based on the inter-comparisons between GIIRS and AGRI for each field-of-view (FOV) and different scene temperatures. The BC method ensures the radiometric consistency between GIIRS and AGRI. Evaluations of GIIRS CCRs show that the mean biases are 0.09, −0.06, and 0.06 K when compared with the three AGRI IR bands, B12, B13, and B14. In addition, BCOCC significantly increases the data yields of successful CCRs by three times that of the optimal cloud-clearing (OCC) approach without BC. For 15 days from September 16–30, 2021, around 37% more GIIRS partially cloudy footprints than clear sky are cloud cleared successfully. The CCRs can be assimilated as CLRs in numerical weather prediction (NWP) models without worrying about the cloud impact. This study provides evidence of the importance of placing an advanced hyperspectral IR sounder and imager onboard the same GEO platform for better quantitative applications.**

***Index Terms*— Advanced Geostationary Radiation Imager (AGRI), bias correction (BC), cloud-cleared radiances (CCRs), Fengyun-4, Geostationary Interferometric Infrared Sounder (GIIRS).**

NOMENCLATURE

GIIRS	Geostationary Interferometric Infrared Sounder.
AGRI	Advanced Geostationary Radiation Imager.
CCR	Cloud-cleared radiance.
CC	Cloud-clearing.
OCC	Optimal cloud-clearing.
BCOCC	Bias corrected OCC.
BC	Bias correction.
FY-4	Fengyun-4.
IR	Infrared.
LWIR	Long-wave infrared.
MWIR	Middle-wave infrared.
VNIR	Visible and near-infrared.
NWP	Numerical weather prediction.
FOV	Field-of-view.
SFOV	Single FOV.
LEO	Low-Earth-orbiting.
GEO	Geostationary.
FOR	Field-of-regard.
CLM	Cloud mask.
DISK	Full disk observation.
REGC	Regional observation area of China.
BT	Brightness temperature.
NSMC	National Satellite Meteorological Center.
CLA	Cloud amount.
CLR	Clear radiance.
SRF	Spectral response function.
VZA	View zenith angle.
STD	Standard deviation.
UI	Uniformity index.
GRAPES	Global/regional assimilation and prediction enhanced system.
CMA	China Meteorological Administration.
QC	Quality control.

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NEdR	Noise equivalent differential radiance.
NEdT	Noise equivalent differential temperature.
CCflag	CC quality flag.
rms	Root mean square.
TBRMS	RMS of BT difference.
BTD	BT difference.
GSICS	Global space-based inter-calibration system.
MTG	Meteosat third generation.
GeoXO	Geostationary extended observations.

I. Introduction

THE Fengyun-4 satellite series introduces a new generation of China's GEO meteorological satellites [1]. The first two satellites, Fengyun-4A (FY-4A) and Fengyun-4B (FY-4B), were successfully launched on December 11, 2016 and June 3, 2021, respectively. Compared to the FengYun-2 series, which has limited imaging capability and lacked a sounder, the FY-4 series has significantly improved China's GEO observing capabilities for weather prediction and environmental monitoring with the GIIRS. As the first hyperspectral IR sounder on a GEO platform, GIIRS provides a high temporal resolution (as high as 15 min) atmospheric sounding capability for targeted regions with rapidly changing weather. GIIRS/FY-4A consists of a 32 $\times$ 4 sensor array plane and offers 1650 spectral channels with a spectral resolution of 0.625 cm^{-1}. The LWIR spectral region (700–1130 cm^{-1}, or 8.85–14.29 μm) is sensitive to atmospheric temperature, while the MWIR spectral region (1650–2250 cm^{-1}, or 4.44–6.06 μm) is sensitive to atmospheric moisture. The spatial resolution of GIIRS/FY-4A is 16 km at nadir [1], [2]. Despite being an experimental satellite, FY-4A has shown overall stable performance and its measurements have been successfully used for weather prediction and applications [3], [4]. The subsequent satellites in the FY-4 series are or will be operational. The FY-4B shares similar observation capabilities with FY-4A but comes with some improvements [5]. For instance, the spatial resolution of GIIRS on FY-4B increases to 12 km at nadir, reducing the likelihood of cloud interference compared to GIIRS/FY-4A. The Advanced Geostationary Radiation Imager (AGRI) is another primary payload on the FY-4A platform. The AGRI has 14 spectral bands with spatial resolutions ranging from 0.5 to 1 km for the VNIR bands and 2–4 km for the IR spectral bands. It offers high radiometric accuracy and provides numerous quantitative products for weather forecasts and environment monitoring, including clouds, atmosphere, surface, and radiation products [6], [7].

Assimilating all-sky radiances in NWP models is desired to fully utilize thermodynamic and hydrometric information provided by IR sounders [8], [9]. Microwave sounder data have been assimilated under all-sky conditions [10]. On the other hand, most operational NWP models only assimilate clear IR radiances from clear FOVs or clear channels unaffected by clouds [2], [3]. This approach significantly reduces the number of observations to be assimilated in regions with extensive cloud cover. However, directly assimilating cloud contaminated IR radiances remains challenging due to difficulties and limitations in modeling clouds in radiative transfer models [11]. As an alternative approach to utilize radiances in cloudy skies, CC techniques provide clear equivalent radiances or CCRs by combining the advanced IR sounder with an advanced microwave sounder [12], [13] or a high spatial resolution imager [14], [15], [16], [17], [18], [19] onboard the same platform. The advantage of the imager-based CC approach compared to the microwave-based one is that the former maintains the original SFOV resolution while the latter degrades to the microwave sounder resolution for CCRs. Maintaining the SFOV resolution is important to effectively assimilate thermodynamic information into NWP models under partially cloudy skies [20], [21].

This study extends previous investigations of the LEO imager-based OCC approach [14], [15], [16], [17], [18], [19] and adapts it to the GEO-based GIIRS using AGRI onboard the same platform of FY-4A. The synergistic use of a hyperspectral IR sounder and an imager onboard the same platform offers many benefits for radiance assimilation. In cloudy skies, the imager can be used to improve the sounder's cloud products and sub-footprint cloud characterization [22]. The inter-calibration and validation also benefit from the same platform observations as both sensors share the same viewing geometry, which allows inter-comparisons with different zenith angles [23], [24], [25], [26]. Similarly, the inter-calibration between GIIRS and AGRI onboard the same GEO platform not only helps with understanding the radiometric differences between the two sensors but also offers the unique opportunity to understand their diurnal characteristics [27]. If the radiometric differences between GIIRS and AGRI are well characterized, the GIIRS CCRs can be derived.

Focusing on a hyperspectral IR sounder from a GEO platform, this research aims to investigate the feasibility of adapting imager-based OCC approaches from LEO to GEO platforms. In comparison to LEO satellites, there are notable gaps between GIIRS/FY-4A footprints, i.e., 16 km between nearby columns of detector arrays and 4.53 km between nearby detectors on the same column array. These gaps increase the distance between two nearby GIIRS FOVs to 32 km in the east/west direction and 20.53 km in the north/south direction. One of the assumptions for CCRs is that adjacent cloudy footprints share the same thermodynamics in both clear and cloudy portions [16]. The larger gaps between GIIRS FOVs on GEO platforms may pose challenges in meeting this assumption, potentially resulting in less successful CCRs. Di et al. [28] demonstrated that such challenges can be partially mitigated by taking advantage of the high temporal resolution, 15 min, offered from the targeted GIIRS observations. However, such high temporal resolution is not always available for GIIRS observations. It is therefore important to explore the GIIRS CCRs using traditional spatial information.

Another important aspect is the radiometric consistency between GIIRS and AGRI, which needs to be characterized and understood. Previous studies have shown a mean bias within 1 K between GIIRS and AGRI for the AGRI window IR bands B12 and B13 [27]. To derive accurate CCRs, it is essential to include at least one absorption band. However, the systematic bias between GIIRS and

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AGRI at the CO_2 absorption band B14 causes less successful CCRs [29]. To make better use of the AGRI CO_2 absorption band B14, further assessments of the radiometric consistency between GIIRS and AGRI are necessary. The inter-comparisons between GIIRS and AGRI should be conducted for all three AGRI IR bands overlapping with the GIIRS spectra and used for CCR generation. Furthermore, Yin et al. [2] showed that the bias between the observation and background (O–B) of GIIRS LWIR depends on FOVs and latitudinal distribution. As a result, more comprehensive inter-comparisons between GIIRS and AGRI are needed, to ensure the radiometric consistency between GIIRS and AGRI for all GIIRS FOVs, which allows imager-based GIIRS CCRs to be accurately derived.

The observation data used for this study are described in Section II. The methodologies for inter-comparison and OCC approaches are presented in Section III. The results and analysis are given in Sections IV and V, respectively, and the summary and discussion are given in Section VI. Acronyms are given in Nomenclature for this article, and the FY-4A observation timetable is given in Appendix.

II. Data

A. Geostationary Interferometric Infrared Sounder

The GIIRS L1b radiances and geolocation data are used. FY-4A GIIRS has a sensor array plane consisting of 32 × 4 square-shaped detectors, also referred to as the FOR. In normal mode, the GIIRS has observed the region of China, from 3° N to 55° N, and 60° E to 137° E, every 2 h, since April 2019. The sensor observes seven scan lines from north to south; each scan line contains 59 FORs (from west to east), taking 15 min. The last 15 min for every 2 h has no observation since the GIIRS sensor needs to turn around from south to north. Besides, the GIIRS has no observation from 1600 to 1900 UTC due to the calibration timetable. As a result, there are at most 11 scans of the regional area per day for GIIRS observation in normal mode [30]. In addition, a more rapid observation mode will be turned on for targeted observing areas when needed, with a higher temporal resolution of 15–30 min. The processing software of GIIRS L1b was updated in August and December 2019, respectively, from version 1 (v1) to version 3 (v3) [30]. For the GIIRS spectrum data, the hamming apodization is used [31].

B. AGRI

The AGRI L1b radiances and geolocation data, as well as the AGRI L2 CLM products, are used. Both the AGRI DISK and the REGC are used for collocation with GIIRS, whichever is closer to GIIRS in time is used for temporal collocation. For FY-4A AGRI, it observes a full DISK for the first 15 min every 1 h, and three full DISKs in a roll every 3 h. Otherwise, the AGRI observes the REGC every 5 min. There is no observation from 1715 to 1730 UTC due to sensor adjustment. As a result, to temporal collocate AGRI to GIIRS, the time difference is less than 15 min when AGRI is observing the DISK, while the time difference is less than 5 min when AGRI is observing the REGC.

Unlike the sounders and imagers onboard the LEO satellites, the FY-4A AGRI and GIIRS have independent observation modes and timetables. In normal mode, GIIRS is a slow scan sensor, while AGRI is a fast scan sensor. The FY-4A GIIRS normal observation mode and the FY-4A GIIRS and AGRI observation timetable are summarized in Appendix. To help understand the data yields of CCRs, 15 days from September 16–30, 2021 are used for inter-comparisons and CCR generation. The data are provided by the NSMC (https://satellite.nsmc.org.cn).

III. Methodologies

A. Collocation

The spatial resolution is 16 km for FY-4A GIIRS and 4 km for AGRI IR bands and CLM products. For each GIIRS footprint, a square box of 5 × 5 collocated AGRI pixels or 20 × 20 km is determined by the center AGRI pixel with the smallest spherical distance from the center location of the GIIRS footprint. The slightly larger coverage of 5 × 5 AGRI pixels is used to collocate GIIRS to account for the diffraction effects [32]. Fig. 1 shows the AGRI band B12 (10.8 μm) BT (K) and CLM products, before (top) and after (bottom) collocating with GIIRS footprints. From north to south (FOVs 1–32) and from west to east (columns 1–4), the 32 × 4 sensor array plane of GIIRS footprints is highlighted with black squares. Note that there are significant gaps between two adjacent GIIRS column FOVs. These gaps make the adjacent GIIRS column FOVs further away from each other and may pose additional difficulty for CCR generation. Since the geolocation accuracy of FY-4A AGRI and GIIRS may significantly affect the CCR estimation, the process of the navigation and registration system is used for determining the observation timetable and geolocation data of instruments on FY-4A [1]. Overall, the geolocation quality or error of band-to-band registration is less than 1/4 pixel [28], which may pose no significant risk for CCR demonstration in this study.

The operational AGRI CLM product [6] has four categories of cloud cover, including "cloudy," "probably cloudy," "probably clear," and "clear." In this study, the "cloudy" and "probably cloudy" are considered as cloudy; and the "clear" and "probably clear" are considered as clear. For each GIIRS footprint, the CLA is defined as the percentage comparing the cloudy pixel numbers to the total collocated pixel numbers of AGRI [22]. The quality flags from both AGRI L1 and L2 products are used to identify valid pixels. If more than 20% of the collocated AGRI pixels have valid values, then the mean radiances of AGRI IR bands and the CLA are calculated by the following equations:

$$R_M^i = \frac{\sum \mathrm{Rad}_i(ipix) * w(ipix)}{\sum w(ipix)} \tag{1}$$

$$R_M^{\mathrm{clr},i} = \frac{\sum \mathrm{Rad}_i(ipix) * w(ipix) * w_{\mathrm{clr}}(ipix)}{\sum w(ipix) * w_{\mathrm{clr}}(ipix)} \tag{2}$$

$$\mathrm{CLA} = 1 - \frac{\sum w(ipix) * w_{\mathrm{clr}}(ipix)}{\sum w(ipix)} \tag{3}$$

where R_M^i (or $R_M^{\mathrm{clr},i}$) is the spatially averaged radiance of the collocated AGRI IR band radiances (or CLRs), and Rad_i

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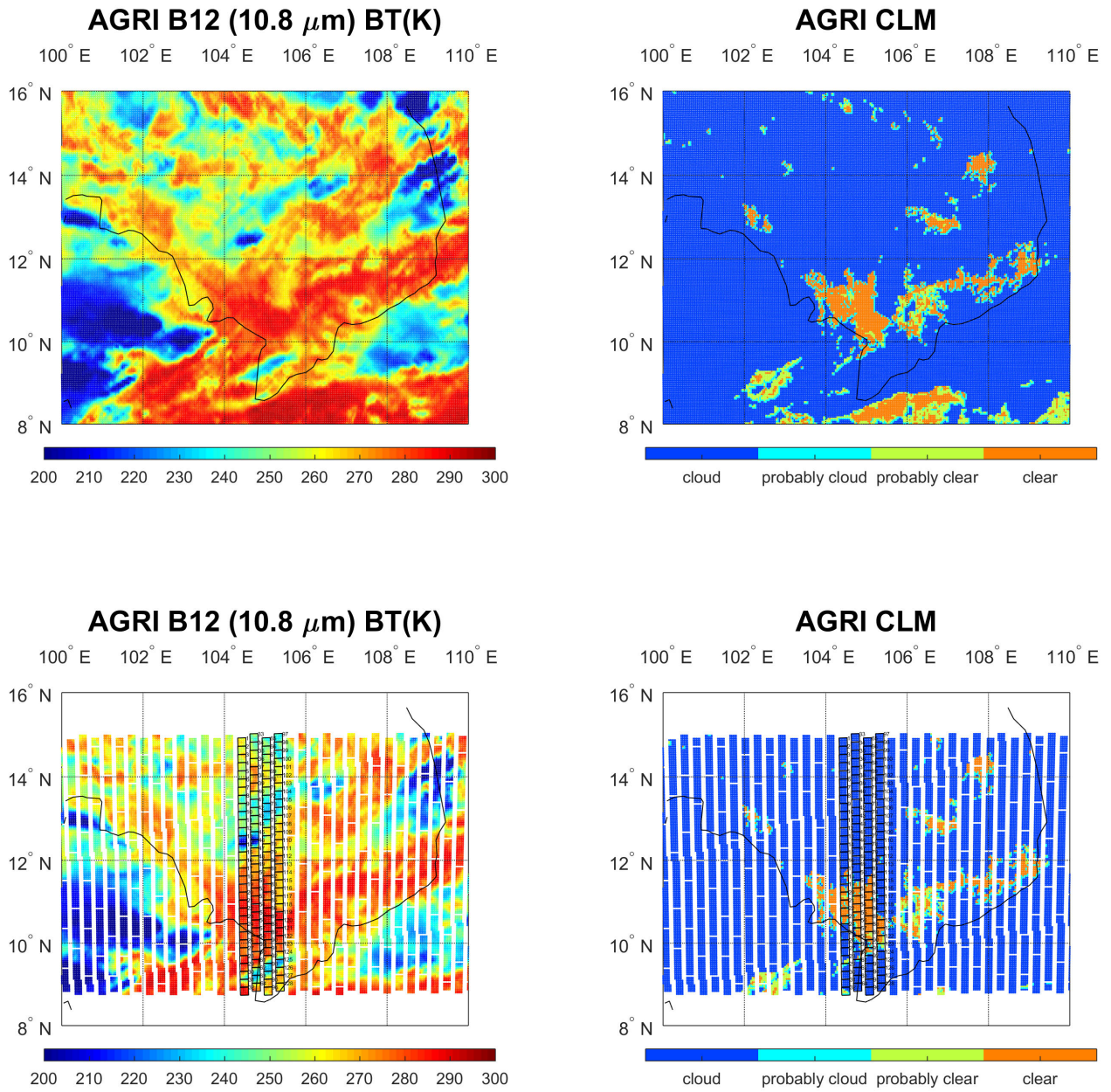


Fig. 1. AGRI B12 (10.8 μm) BT (K) and CLM (top) before and (bottom) after collocating with GIIRS footprints. From north to south (FOV 1–32) and from west to east (column 1–4), the 32 $\times$ 4 sensor array plane of GIIRS footprints is highlighted with black squares.

is the AGRI IR radiance for band i. For AGRI pixel $ipix$, if it is successfully collocated with the GIIRS footprint, $w(ipix) = 1$, otherwise, $w(ipix) = 0$. And if it is a clear pixel, $w_{\text{clr}}(ipix) = 1$; otherwise, $w_{\text{clr}}(ipix) = 0$.

Fig. 2 shows the SRFs of AGRI IR bands overlapping with one randomly chosen GIIRS BT spectrum. Only the AGRI IR bands B12 (10.8 μm), B13 (12.0 μm), and B14 (13.5 μm) have full spectral overlaps with the GIIRS spectrum. These three AGRI IR bands (two window and one CO_2 absorption) are suitable and will be used for inter-comparisons and GIIRS CCR generations. Note that the GIIRS water vapor band (1650–2250 cm^{-1}) is on the shortwave side of the 6.5 μm water vapor absorption band. It does not have a full spectral overlap with the two AGRI water vapor absorption bands. In addition, the GIIRS CCRs are only generated for the GIIRS LWIR spectrum, due to the lack of overlap between the AGRI water vapor absorption bands and the GIIRS MWIR spectrum. The GIIRS radiances are spectrally convolved with the AGRI IR bands' SRFs by

$$f_i(R_\upsilon) = \frac{\int R_\upsilon \times \text{SRF}_i(\upsilon) d\upsilon}{\int \text{SRF}_i(\upsilon) d\upsilon} \tag{4}$$

where $f_i(R_\upsilon)$ is the spectrally convolved radiance of GIIRS for AGRI band i, and R_υ is the GIIRS radiance of wavenumber υ.

B. Inter-Comparisons

The GIIRS/AGRI matched-up samples are divided into different categories, according to scene temperature, VZA, latitude, and GIIRS FOV number. The convolved radiances of GIIRS are inter-compared to the mean radiances of AGRI.

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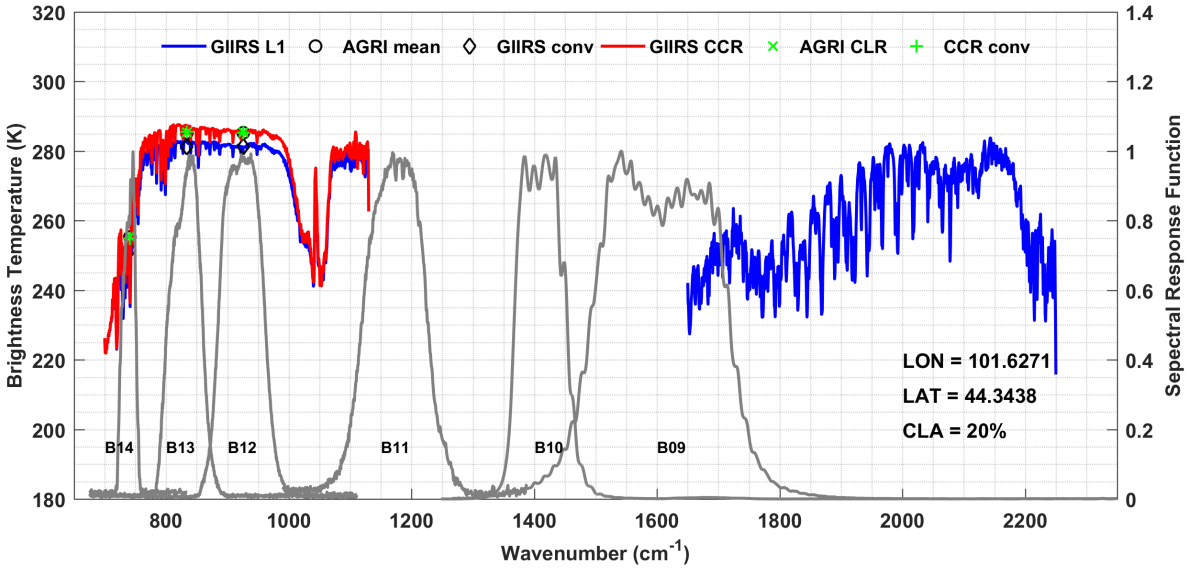


Fig. 2. SRFs (gray lines) of AGRI IR bands overlapping with one randomly chosen BT (K) spectrum converted from the original GIIRS radiances (blue line), as well as the CCRs (red line) of the GIIRS long-wavelength IR spectrum. Only the AGRI IR bands B12 (10.8 μm), B13 (12.0 μm), and B14 (13.5 μm) having full spectral overlaps with the GIIRS spectrum are used for inter-comparison and CCR generations. The convolved radiances of GIIRS (GIIRS conv, black ◇) and CCRs (CCR conv, green +), as well as the mean radiances of AGRI (AGRI mean, black ◯) and AGRI CLRs (green ×), for the three AGRI IR bands B12, B13, and B14 are shown. The CCR conv "+" and AGRI CLR "×" are overlapped and shown as green star symbols, indicating the radiometric consistency of the GIIRS CCRs and AGRI CLRs for the three AGRI IR bands.

The mean bias and STD of radiance differences of (AGRI–GIIRS) are calculated for each AGRI IR band and for each category. The mean bias represents the radiometric consistency of GIIRS observation inter-compared to AGRI, while the STD represents the uncertainty of the radiometric consistency. The mean bias and STD calculations are done in the radiance domain, and the radiance bias and STD are converted to BT using the differential Planck function for better viewing and easier understanding.

Around 52 864 GIIRS footprints are collocated with AGRI pixels every 2 h. From a half month of observations, around 8 million GIIRS/AGRI matched-up samples are prepared for inter-comparisons and GIIRS CCR generations. To avoid the potential uncertainties caused by geolocation errors and the collocation method, the UI is defined as the ratio of STD to mean BT of the collocated AGRI pixels [23], [24], [25], [26]. STD is also calculated in the BT domain. The AGRI window IR band B12 (10.8 μm) is the cleanest band of AGRI. Therefore, the AGRI band B12 (10.8 μm) is used to calculate the UI for each GIIRS footprint. The smaller the UI threshold, the more uniform the scenes are. With a UI threshold of 0.01, more than 99% of clear skies pass the uniformity test, while only one-third of overcast skies pass the test. In this study, only clear sky footprints with a UI less than 0.01 are used for inter-comparisons.

In addition, the GIIRS footprints with VZAs larger than 60° are removed for analysis, as large VZAs with large footprint size bring large uncertainties in geolocation errors and are abandoned for data assimilation in NWP models [2]. To evaluate the GIIRS LWIR radiances in the GRAPES of the CMA, an absolute bias threshold of 3 K in the observation and background differences (O−B) of the GIIRS window channel 362 (10.8 μm) is used as another QC for clear sky detection. This scheme flagged approximately 4% of clear observations [2]. Similarly, an absolute bias threshold of 5 K of (AGRI–GIIRS) for the cleanest AGRI window band B12 is used to eliminate the observation outliers. This additional QC eliminates around 1% of all clear observations. As a result, more than 1.1 million GIIRS clear sky footprints passing all above QCs are used for inter-comparisons. The abundance of clear observations ensures reliable assessments of the radiometric consistency between GIIRS and AGRI.

C. Optimal Cloud-Clearing

In order to remove cloud effects, homogeneousness is assumed in both clear and cloudy portions for the two adjacent GIIRS footprints. This is the basic assumption of the OCC approach [16]—that the radiance difference between the two adjacent footprints is solely caused by the difference in cloud cover [14]. The OCC approach for atmospheric infrared sounder (AIRS) radiances uses nine moderate resolution imaging spectroradiometer (MODIS) IR bands, including four CO_2 and two H_2O absorption bands [16]. The CCRs are estimated by

$$R_v^{\mathrm{cc}} = \frac{R_v^1 - R_v^2 N^*}{1 - N^*} \tag{5}$$

where R_v^{cc} is the CCR of wavenumber v; R_v^1 and R_v^2 are the cloudy radiances from the two adjacent footprints; and N^* [33], [34] is the ratio of the effective CLA between the two adjacent footprints calculated by

$$N^* = \frac{\sum_i \left\{ \left[f_i\left(R_v^1\right) - R_M^{\mathrm{clr},i} \right] \times \left[f_i\left(R_v^1\right) - f_i\left(R_v^2\right) \right] / \mathrm{NEdR}_i^2 \right\}}{\sum_i \left\{ \left[f_i\left(R_v^2\right) - R_M^{\mathrm{clr},i} \right] \times \left[f_i\left(R_v^1\right) - f_i\left(R_v^2\right) \right] / \mathrm{NEdR}_i^2 \right\}} \tag{6}$$

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where $f_i(R_v)$ is the GIIRS convolved radiances for imager AGRI band i, and $R_M^{\mathrm{clr},i}$ is the averaged AGRI CLRs collocated with the two adjacent GIIRS footprints. The NEdR_i is the noise equivalent differential radiance converted from the NEdT (0.2, 0.2, and 0.5 K at a scene temperature of 300 K, for AGRI bands B12, B13, and B14, respectively [1]). According to (5), when either the principal or secondary FOV is a clear sky, the derived CCR for the other partially cloudy FOV will be exactly the same as the radiances of the clear sky FOV, providing no unique information. Therefore, the CC procedure is only processed between two adjacent FOVs that are both partially cloudy.

For each GIIRS footprint, based on the cloud condition and the QCs of CC results, a CCflag is assigned. The imager CLRs are used in two ways to help: 1) calculate the value of N^* and 2) QCs of the CCRs. Due to the lack of overlap between AGRI absorption bands and the GIIRS radiance spectrum, two additional QCs are used to remove those CCRs with low quality [18]. The first QC, or the homogeneous test, examines if the pair of GIIRS footprints used violate the homogeneous assumption. Since the radiance difference between the pair of footprints is solely caused by cloud cover difference, the GIIRS footprint with less cloud cover should be warmer than the other one. Violation of this assumption indicates that the footprint pair is not suitable for CCR generation. The second QC is based on the similarity test. Equation (5) shows that the pair of footprints have to be substantially different in cloud cover. Otherwise, N^* would be close to 1, which means the pair of footprints provide little additional information compared with each other, making it challenging to derive the CCRs. The CCRs that violate the homogeneous or similarity tests are rejected and marked as CC fail (CCflag = 1). However, the additional QCs do have limitations over the temperature inversion regions, where less cloudy radiances are not necessarily warmer than cloudy radiances, such as maritime stratus clouds near cold ocean and subsidence regions. This is less of a problem for AIRS/MODIS, since nine MODIS IR spectral bands, including four CO_2 and two H_2O absorption bands, are used to generate the AIRS CCRs [16]. As for cross-track infrared sounder (CrIS)/visible infrared imaging radiometer suite (VIIRS) or GIIRS/AGRI, the additional QCs are necessary to remove those CCRs with low qualities [17], [18], [19], [28], [29], although at the cost of the potential to reject good CCRs over temperature inversion region.

Successful CCRs could be assimilated as CLRs in NWP models. However, the CCRs estimated by (5) should have more observational noise than the GIIRS original cloudy radiances [19]. The observation noise in CCRs is inflated according to

$$\gamma = \sqrt{\frac{1+N^{*2}}{(1-N^*)^2}}. \tag{7}$$

Therefore the CCR noise amplification is highly sensitive to N^*. When N^* is close to 1, it indicates that the two adjacent footprints are highly similar to each other, and the CCR noise is significantly amplified. When N^* is 0, it becomes the extreme case of a clear sky situation. To avoid highly amplified observation errors, N^* is required to be either smaller than 0.25 or larger than 4, making the noise amplification factor between 1 and 1.37 [19]. Assimilating the CCRs as CLRs has shown a positive impact on improving high-impact weather forecasts [17], [18]. Appropriately handling the inflated observation errors for CCRs has proven a further positive impact on hurricane forecasts [19].

In addition, the CCRs should be close to the imager CLRs. The TBRMS of the three bands used is calculated by

$$\mathrm{TBRMS} = \sqrt{\sum_i \left[T_b\left(R_M^{\mathrm{clr},i}\right) - T_b\left(f_i\left(R_v^{\mathrm{cc}}\right)\right)\right]^2 / I} \tag{8}$$

where T_b is the function converting radiance to BT for each AGRI band; and I is the total number of AGRI IR bands used for CCR generation, which is three for a GIIRS/AGRI pair. A threshold of 0.5 K of TBRMS was used for selecting successful AIRS CCRs using nine bands of MODIS, since AIRS and MODIS have very good agreement with each other. However, a TBRMS threshold of 1 K is used for GIIRS CCRs, which is around twice the rms of the three AGRI bands' NEdTs at mean scene temperature for clear sky only. If the TBRMS is greater than the threshold, the GIIRS CCRs are rejected and marked as CC fail (CCflag = 2); otherwise, they are marked as CC success (CCflag = 3). Furthermore, the high-confidence CCRs (CCflag = 4) are identified if the absolute BTDs between the convolved radiances of CCRs and AGRI CLRs of the principal footprint are smaller than the given thresholds for the three AGRI IR bands. The BTD thresholds are equal to the NEdTs at mean scene temperature for clear sky only, which are 0.2, 0.2, and 0.7 K for AGRI bands B12, B13, and B14, respectively.

D. Bias-Corrected OCC

The observation errors may come from the instrument or the calibration process. If the radiometric differences between the imager and sounder are well characterized, the imager-based CCRs can be derived with good quality [16], [18]. To generate accurate GIIRS CCRs, the GIIRS and AGRI should have similar radiometric accuracy. Therefore, a BC scheme based on the inter-comparisons between GIIRS and AGRI for clear sky is introduced by

$$R_{\mathrm{BC}}^i = R_M^i - dR_M^i \tag{9}$$

where R_M^i and R_{BC}^i are the mean radiances of AGRI before and after applying the radiance BC, and dR_M^i is the radiance bias for each AGRI band i. This needs to be done for different scene temperatures and each FOV because each of them may have a different radiance bias.

To help distinguish different CC methods, the OCC method refers to the original CC method without radiance BC, while the BCOCC method uses the bias-corrected radiances of AGRI CLRs to help generate the GIIRS CCRs. OCC is suitable for duo sensors with great radiometric consistency such as AIRS/MODIS [16] or CrIS/VIIRS [18]. The two additional QCs based on the homogenous test and similarity test are used for both OCC and BCOCC methods to remove the CCRs with low quality. The only difference is that BCOCC uses the bias-corrected AGRI CLRs to generate the GIIRS CCRs. And the BC scheme affects the CC results in three ways: 1) to

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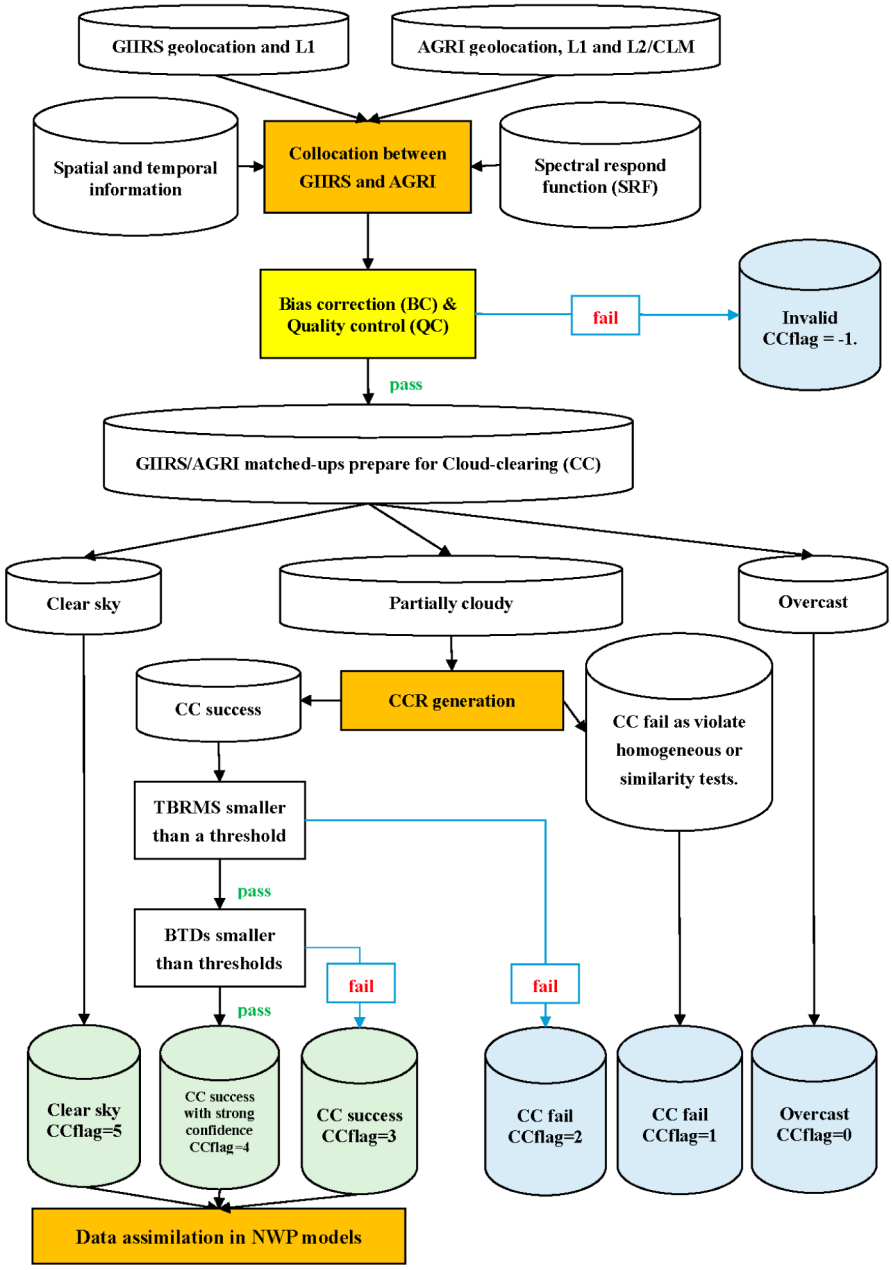


Fig. 3. Flowchart for GIIRS/AGRI CCR generation.

calculate the value of N^*; 2) QCs of the CCRs; and 3) the CCR noise amplification γ is highly related to N^*. Fig. 3 shows the flowchart of CCR generation.

IV. Inter-Comparisons

A. Scene Temperature, VZA, and Latitude

Fig. 4 shows the mean biases and STDs of (AGRI–GIIRS) according to (a) scene temperature, (b) VZA, (c) latitude, and (d)–(f) FOV number for AGRI band B12, B13, and B14, respectively, before applying BC. All processing is in the radiance domain except for visualization where the radiance bias and STD are converted to BT.

From Fig. 4(a), the mean bias and STD of (AGRI–GIIRS) are highly dependent on the scene temperature. For AGRI window IR bands B12 and B13, mean biases are −0.07 and −0.02 K, for scene temperatures between 280 and 300 K. More than 80% of all GIIRS clear sky footprints fall into this group. Furthermore, when the scene temperature is colder than 280 K (or warmer than 300 K), the AGRI radiances are colder (or warmer) than the convolved GIIRS radiances. The colder (or warmer) the scene temperature, the more negative (or positive) the radiance biases for the two window IR bands. Of the clear samples, less than 0.2% have a scene temperature warmer than 320 K or colder than 260 K. However, this 0.2% of clear samples have a larger absolute radiance bias along with larger uncertainty in the bias as seen from the STD. These findings are consistent with previous studies [27]. Those clear sky footprints with extremely cold scene temperatures

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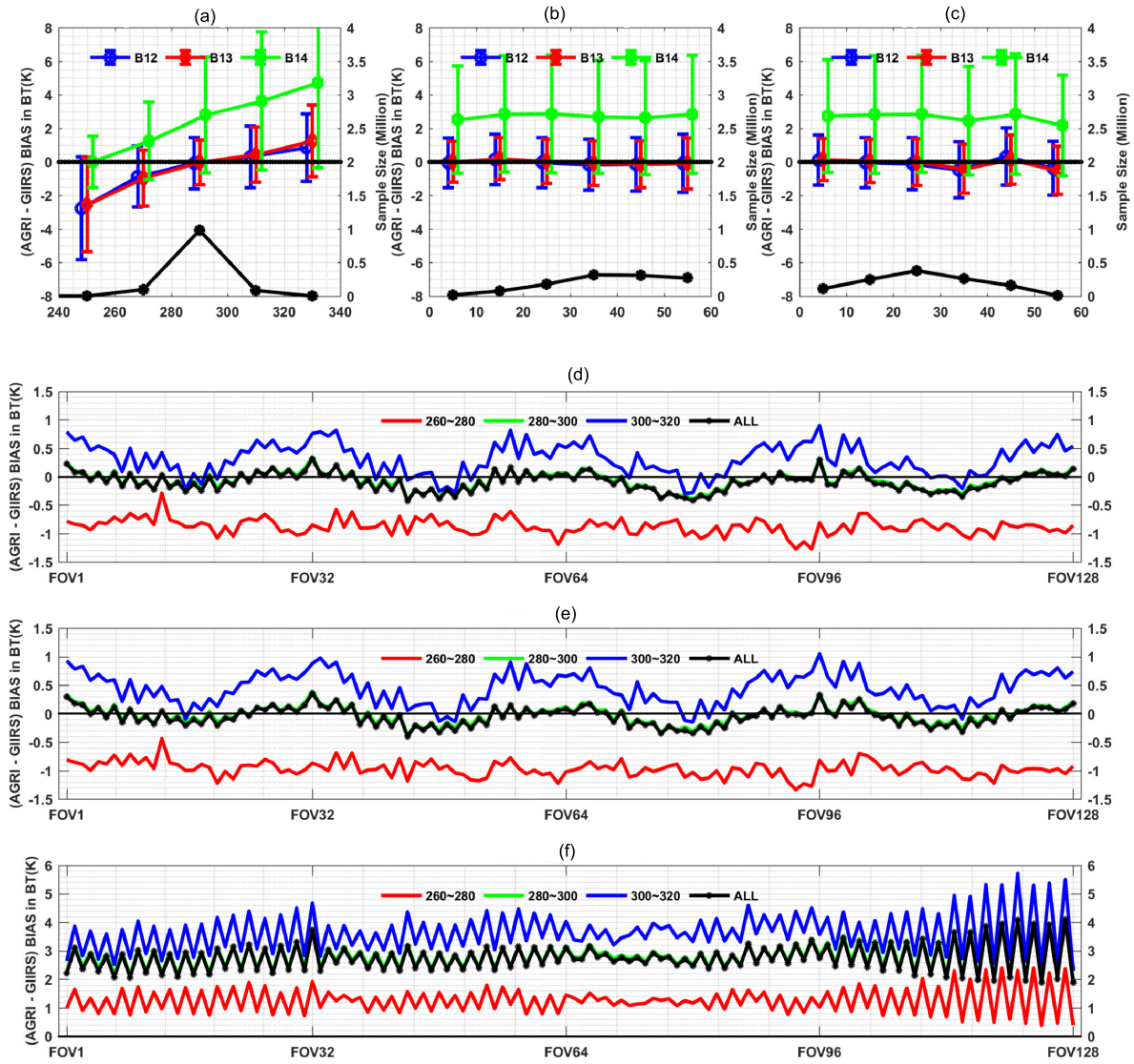


Fig. 4. Mean bias (colored line) and STD (bar) in BT (K) of (AGRI–GIIRS) and the sample size (million) (black bold line), according to (a) scene temperature, (b) VZA, and (c) latitude, as well as the mean bias (colored line) of (AGRI–GIIRS) according to FOV for the AGRI IR bands (d) B12, (e) B13, and (f) B14, respectively, before applying the radiance BC. The mean bias and STD calculations are processed in the radiance domain, and the radiance bias and STD are converted to BT using the differential Plank function for visualization. Note that, the green line is overlapped with the black line in (d)–(f), since most of the clear sky samples have scene temperatures between 280 and 300 K.

are mostly located around the north edge of the Tibetan Plateau. Only around 200 footprints out of 1.1 million clear sky samples have scene temperatures less than 260 K, and the mean scene temperature for this group is 258.5 K. As for AGRI CO_2 absorption band B14, the mean biases are 2–4 K warmer than for the two window IR bands, for all scene temperatures. These radiance biases between AGRI and GIIRS, if uncorrected, will introduce additional errors in the GIIRS CCRs, and thus must be corrected.

From Fig. 4(b), the mean bias and STD of (AGRI–GIIRS) have a subtle dependence on VZA. For AGRI window IR bands B12 and B13, the two sensors have overall good agreement with each other across different VZAs. The absolute mean biases of (AGRI–GIIRS) are close to zero, with the maximum being less than 0.2 K for a VZA less than 60°. However, they do show a shift from positive (a maximum of 0.19 K) to negative (a minimum of −0.16 K) when VZAs increase from 0°–30° to 30°–60°. These results are consistent with the scene temperature dependence from Fig. 4(a). Statistically, larger VZAs are associated with larger optical depth. In addition, large VZAs for GEO satellites may come from the mid-latitudes and higher, while small VZAs exclusively come from the tropics. As a result, scene radiances with larger VZAs are statistically colder than those with smaller VZAs and tend to have negative radiance biases. These results are consistent with a previous study [2] showing a linear relationship between the cosine of VZA and O−B for GIIRS LWIR radiances.

From Fig. 4(c), the latitude has a weak impact on the mean bias and STD of (AGRI–GIIRS). For AGRI window IR bands B12 and B13, the absolute mean biases are close to zero with a maximum of 0.14 K for latitudes between 0° and 30° N, and 0.5 K for latitudes between 30° N and 60° N. Over the

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REGC, lower latitudes consist of more ocean surface than higher latitudes. As a result, compared to higher latitudes, lower latitudes have more uniform and warm surfaces and the mean biases are relatively low and flat. On the other hand, the mean bias and STD have larger variations in high latitudes than in low latitudes. Especially for latitudes between 30° N and 40° N, the mean biases are as low as −0.4 K for AGRI window IR bands B12 and B13, due to the low scene temperature of the Tibetan Plateau. A previous study [2] has also pointed out that the observation error for the GIIRS LWIR radiance is more positive in high latitudes than in low latitudes.

B. GIIRS FOV Number

As mentioned before, the detector array or FOR of GIIRS onboard FY-4A has four columns of detectors, and each column has 32 squared detectors or FOVs. Each FOV corresponds to a specific detector and has a spatial horizontal resolution of around 16 km [1], [2]. Fig. 4(d)–(f) shows the mean biases according to scene temperature and FOV number for AGRI bands B12, B13, and B14, respectively. For clear sky only, over 99.8% of all GIIRS footprints have scene temperatures between 260 and 320 K, which are categorized into three groups with a scene temperature interval of 20 K. Warmer (>320 K) or colder (<260 K) scenes are rare and not shown.

For AGRI window IR bands B12 and B13, the mean biases are within ±0.5 K for all FOVs and for all clear samples (black line) in Fig. 4(d) and (e). The mean biases decrease when approaching the center of each detector column (FOV 16, 48, 80, and 112) and increase when approaching the north and south ends of each column (FOV 1, 32, 64, 96, and 128). These results indicate that the mean radiances of AGRI are slightly colder than those of the GIIRS for the center FOVs and slightly warmer for the FOVs near the two ends. This is likely due to reduced calibration accuracy near the south and north ends of the GIIRS FOR [2]. The last FOVs of the first and third detector columns (FOV 32 and 96) appear to have larger mean biases than all the other FOVs. These findings are consistent with a previous study [2] that the biases of (O–B) for FOV 32 and 96 are larger than others. In addition, a bias oscillation between the odd and even FOVs is more evident for the first and second detector columns than the third and fourth columns, the reason for which is still unclear.

Furthermore, the scene temperature dependence is more obvious for FOVs near the two ends than FOVs near the center of each detector column. Since more than 80% of clear skies have scene temperatures of 280–300 K (green line), they have very similar patterns as all the clear samples (black line). The warmer (or colder) the scene temperature, the more positive (or negative) the mean biases for all FOVs. For scene temperatures warmer than 300 K (blue line), the mean biases of (AGRI–GIIRS) are 0–0.5 K warmer than those of 280–300 K. As a result, the mean radiances of AGRI are warmer than the GIIRS with a maximum within 1 K for FOVs near the two ends of each detector column, and close to the GIIRS for the center FOVs. On the other hand, for scene temperatures colder than 280 K (red line), the mean biases of (AGRI–GIIRS) are colder than those for 280–300 K. And the AGRI are 0.3–1.3 K colder than the GIIRS for all FOVs.

For AGRI CO_2 absorption band B14, the mean radiances of AGRI are 2–4 K warmer than the convolved radiances of GIIRS for all FOVs, for all the clear samples (black line) in Fig. 4(f). And the mean bias of (AGRI–GIIRS) has a dependence on scene temperature and FOV. The warmer the scene temperature, the larger the mean biases for all FOVs. In addition, an obvious bias oscillation between the odd and even FOV is evident. This oscillation is more evident for FOVs of detector columns at the two sides of the GIIRS FOR than the FOVs of detector columns at the center. Furthermore, the oscillation is much weaker for FOV 65–80, near the north side of the third column, than for other FOVs. Based on the personal communication with sensor scientists from NSMC, the large even/odd oscillation may be due to sensor issue. Reduced radiometric calibration accuracies are found for detectors away from the center of GIIRS FOR, especially for detectors locating at the south and north ends [2]. However, the exact cause for this kind of odd/even oscillation bias is still unknown.

Based on the inter-comparisons between GIIRS and AGRI for clear sky, the radiance bias is mainly dependent on the scene temperature and FOV number. The relationship between the radiance bias and the mean scene radiance is filtered using the spline fitting function. A look-up table is developed for different AGRI scene radiances of each FOV number. When performing the radiance BC, the scene radiance and FOV number are used to find the radiance bias for each AGRI band, which is removed from the mean radiance of AGRI.

C. Impact of BC Method

Fig. 5 shows the mean biases and STDs of (AGRI–GIIRS) according to (a) scene temperature, (b) VZA, (c) latitude, and (d)–(f) FOV number for AGRI bands B12, B13, and B14, respectively, after applying BC.

For all clear samples, the residual biases are 0.002, 0.003, and 0.015 K for AGRI bands B12, B13, and B14, respectively. The residual biases are not zero because the BC is performed in the radiance domain and only converted to BT for visualization. From Fig. 5(a), for the three AGRI IR bands, the scene temperature has a very subtle impact on the residual bias and its STD. For scene temperatures between 260 and 320 K, the absolute residual biases between AGRI and GIIRS are mostly close to zero with a maximum within 0.03 K for the three AGRI bands. For scene temperatures colder than 260 K or warmer than 320 K, larger residual biases are found, due to the nonlinearity of the Planck function. Moreover, the VZA and latitudinal dependence of the residual biases are still subtle.

For the two window IR bands B12 and B13, the residual biases are within ±0.03 K for all FOVs and for all clear samples (black line) in Fig. 5(d) and (e). Furthermore, for all scene temperatures, the residual biases are within ±0.1 K for all FOVs. The warmer (or colder) the scene temperature, the more positive (or negative) the residual radiance biases. These results indicate that the BC successfully removes most of the radiance biases between AGRI and GIIRS. For the AGRI CO_2 absorption band B14, the absolute residual bias is within 0.04 K for all FOVs, for all clear samples (black line)

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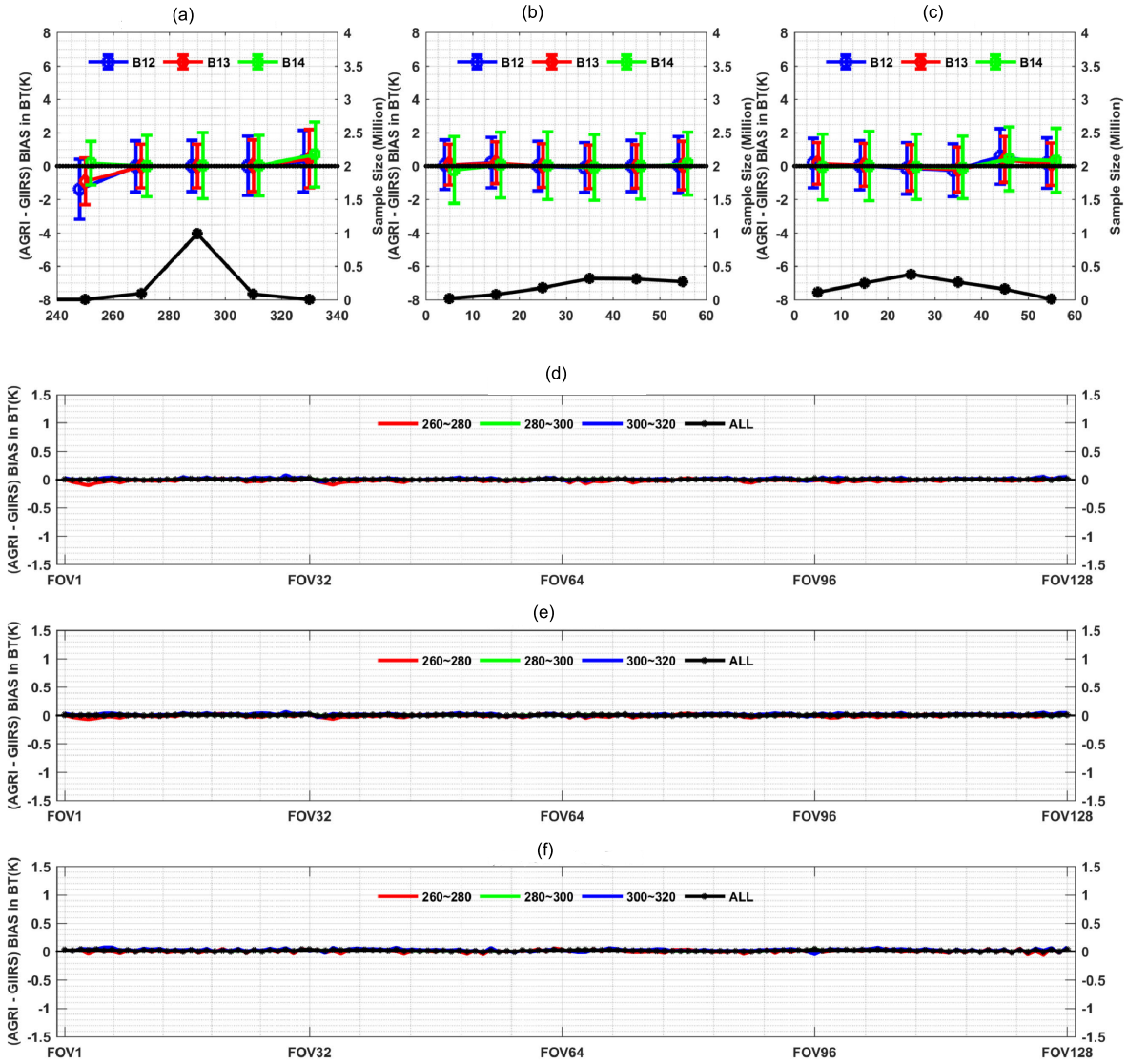


Fig. 5. Mean bias (colored line) and standard deviation (STD, bar) in BT (K) of (AGRI - GIIRS) and the sample size (million) (black bold line), according to (a) scene temperature, (b) view zenith angle, and (c) latitude, as well as the mean bias (colored line) of (AGRI - GIIRS) according to FOV for the AGRI IR bands (d) B12, (e) B13 and (f) B14, respectively, after applying the radiance bias correction. The colored lines for different scene temperatures are overlapped with the black line in (d), (e) and (f), since the mean biases are close to zero after applying the radiance bias correction.

in Fig. 5(f). The absolute residual biases are within 0.08 K, for colder or warmer scene temperatures. Furthermore, the odd/even bias oscillation is largely reduced by the BC method. As a result, after applying the radiance BC, the residual biases for all FOVs are close to zero and much smaller than the NEdTs at mean scene temperature for clear sky. The BC method ensures the radiometric consistency between AGRI and GIIRS, for all the three AGRI IR bands, including the CO_2 absorption band B14.

V. Cloud-Cleared Radiances

The GIIRS CCRs are generated using both OCC and BCOCC methods. The impact of the radiance BC on the data yields and the CCR accuracy is analyzed.

A. Impact on Data Yields

Fig. 6 shows the GIIRS 10.8 μm (925 cm^{-1}) BTs (K) for all sky, clear only, clear and CCRs (CLR + CCR), and the CCflag for the REGC 0000–0145 UTC on September 19, 2021. Observations with VZAs larger than 60° are ignored. The successfully retrieved CCRs partially fill in the gaps between the clear skies and overcast skies. Compared to the OCC method, BCOCC has a more significant impact on the data yields of CCRs. More partially cloudy footprints are cloud-cleared successfully by BCOCC than with the OCC method. The spatial coverages of CLR + CCR from BCOCC [Fig. 6(f)] are significantly larger than those from OCC [Fig. 6(d)]. As a result, BCOCC further increases the number of GIIRS observations that can be assimilated in NWP models without worry of cloud impacts [17], [18].

Fig. 7 shows the number of clear sky and CCRs from the OCC and BCOCC methods, for the 15 days from September 16–30, 2021. More than 1.1 million samples are identified as clear sky. After applying the BC method, around 0.1% of GIIRS clear samples are further removed by QC with a B12 bias difference larger than 5 K. These clear samples likely are

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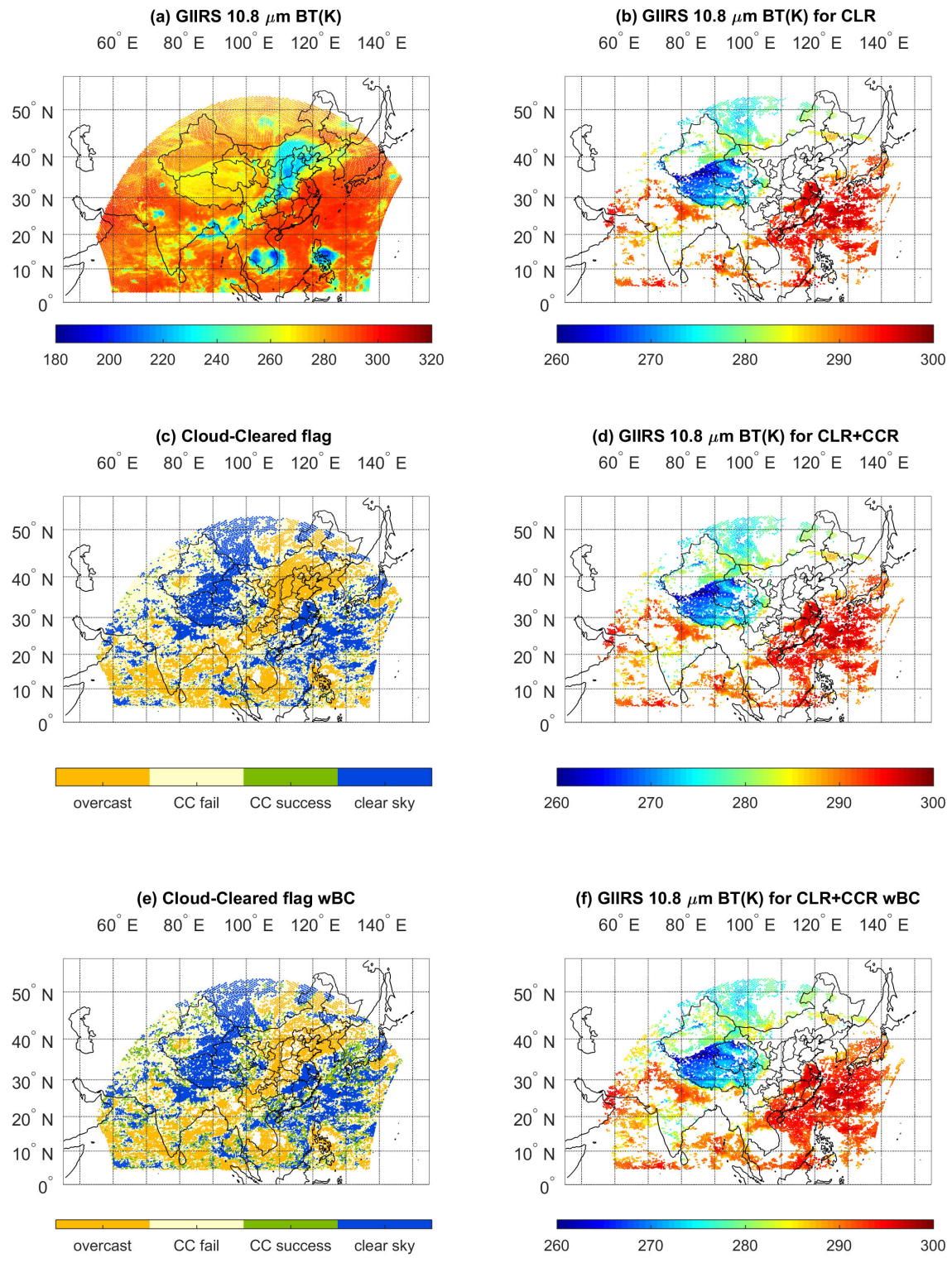


Fig. 6. GIIRS 10.8 μm (925 cm^{-1}) BTs (K) for (a) all sky, (b) clear sky only (CLR), (d) and (f) clear and CCRs (CLR + CCR), and (c) and (e) cloud-cleared flag (CCflag) for the REGC 0000–0145 UTC on September 19, 2021. (c) and (d) Use OCC, while (e) and (f) use BCOCC. Observations with VZAs larger than 60° are ignored. Both the low-confidence CCRs (CCflag = 3) and high-confidence CCRs (CCflag = 4) are CC success (green dots) in (c) and (e).

lower in quality and should be removed from use in NWP models. With the OCC method, only around 12% successful CCRs (138 960 samples) compared to clear observations are made available for potential radiance assimilation in NWP models. As a comparison, successful CCRs from BCOCC (432 183 samples) increase the number of GIIRS observations that can be potentially assimilated as clear observations by 37%, which significantly improves the data yields by more than three times the CCRs from OCC. In total, the successful CCRs from BCOCC and clear sky consist of 23% of all the GIIRS observations, which is significantly more than 17% from the clear sky only. In summary, compared to the OCC

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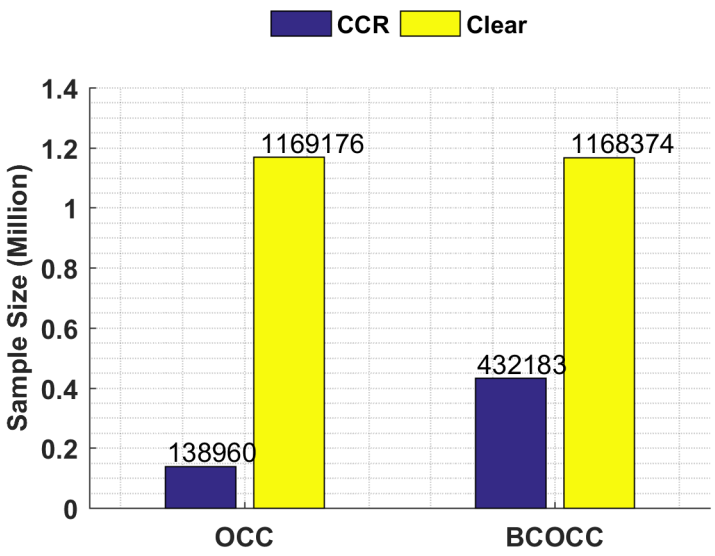


Fig. 7. Sample sizes of clear sky and CCRs from the OCC and BCOCC methods, for 15 days from September 16–30, 2021.

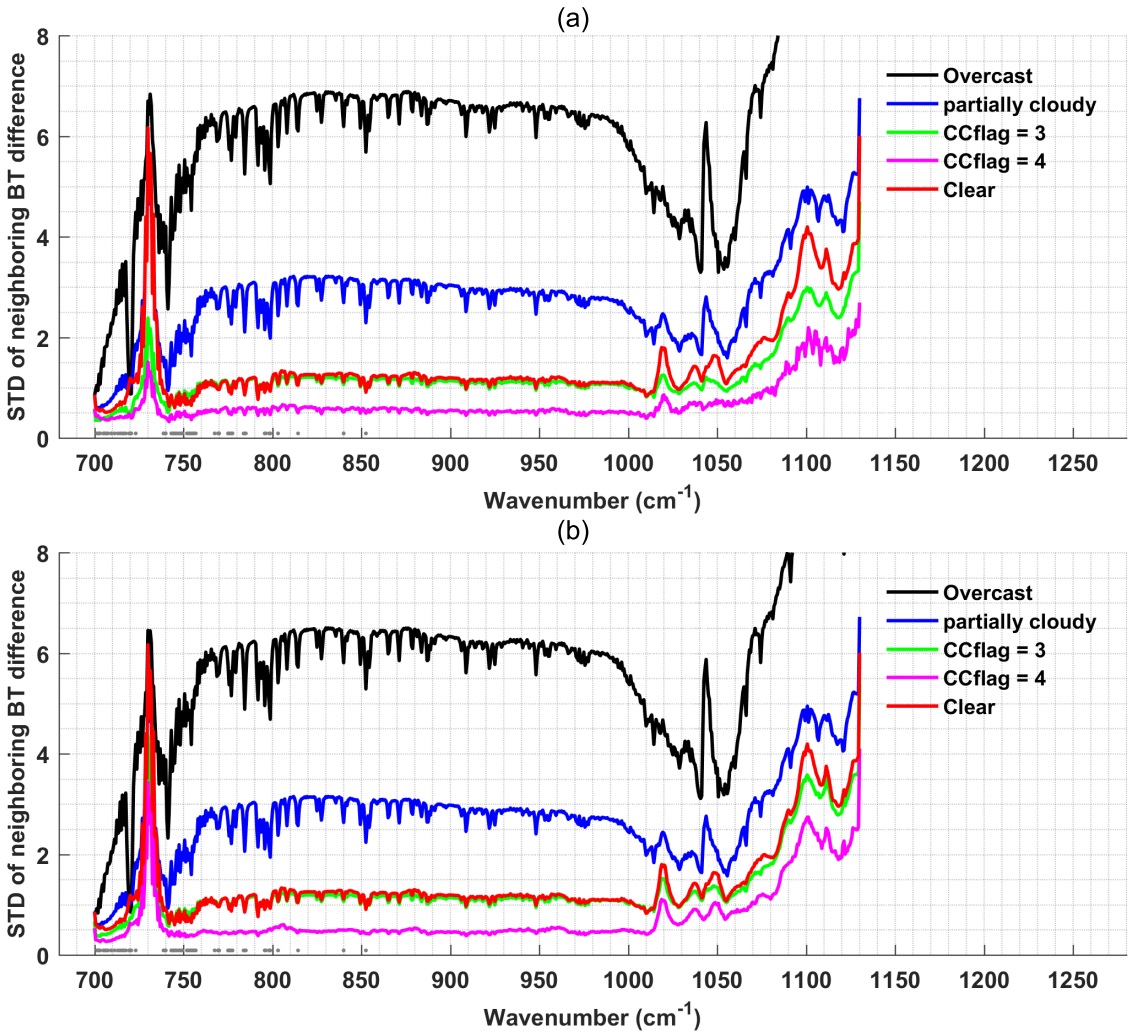


Fig. 8. STDs of neighboring GIIRS footprint differences in BT (K) for CCRs from (a) OCC and (b) BCOCC methods, for September 16–30, 2021. The overcast, partially cloudy, and clear GIIRS footprints are identified from the collocated AGRI CLM. CCflag = 3 (or 4) are successful GIIRS CCRs with lower (or higher) qualities. The 60 GIIRS sounding channels selected for operational data assimilation in CMA's GRAPES are highlighted with gray dots at the bottom.

method, the BCOCC significantly increases the data yields of GIIRS observations that are available to be assimilated into NWP models without worrying about the impact of clouds. As a result, the original imager-based OCC approach

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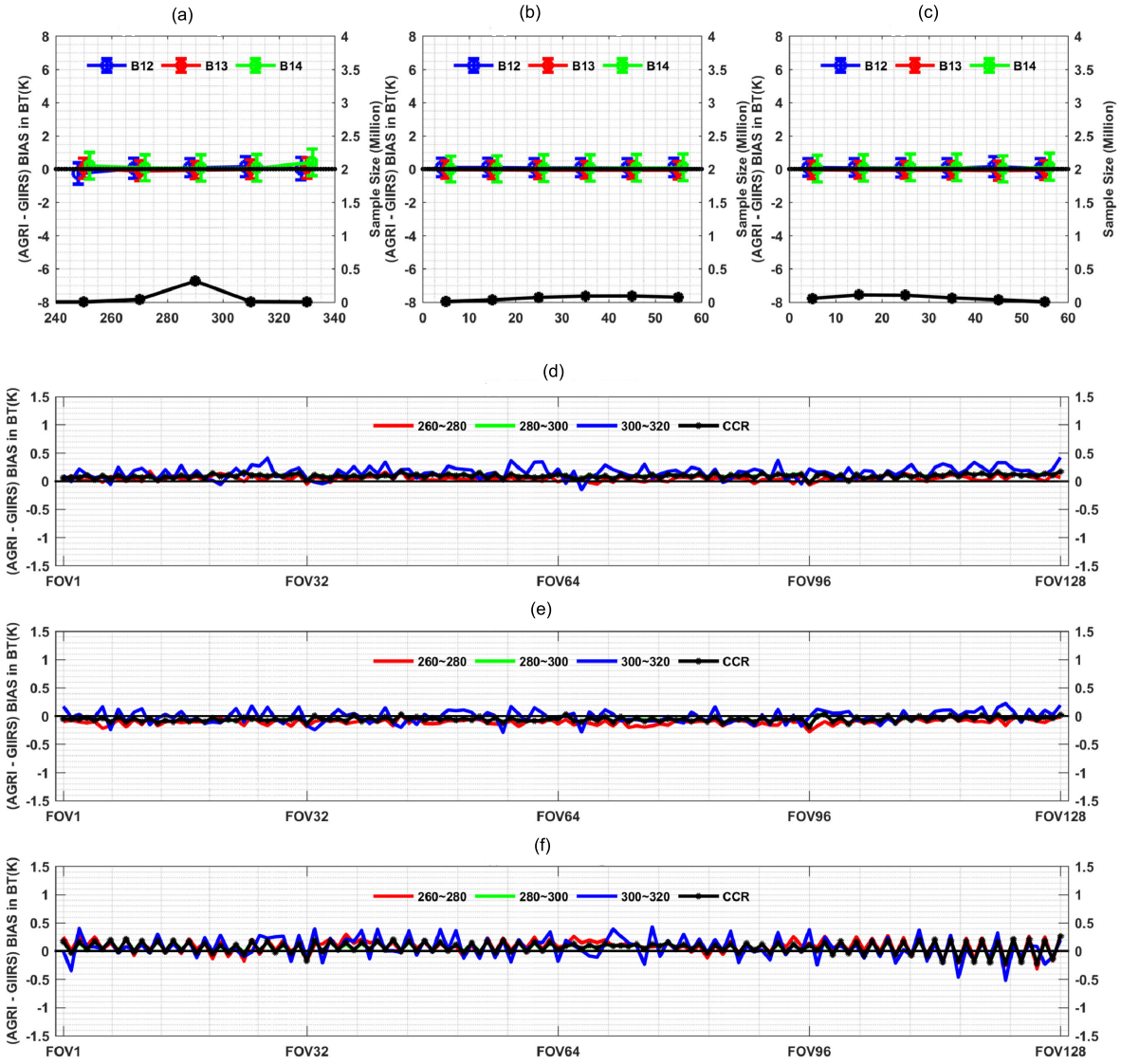


Fig. 9. Mean bias (colored line) and STD (bar) in BT (K) of (AGRI–GIIRS) and the sample size (million) (black bold line), according to (a) scene temperature, (b) VZA, and (c) latitude, as well as the mean bias (colored line) of (AGRI–GIIRS) according to FOV for the AGRI IR bands (d) B12, (e) B13, and (f) B14, respectively, for the CCRs from BCOCC. The mean bias and STD calculations are processed in the radiance domain, and the radiance bias and STD are converted to BT using the differential Plank function for visualization. Note that, the green line is overlapped with the black line in (d)–(f) since most of the CCRs have scene temperatures between 280 and 300 K.

is only suitable for duo sensors with great radiometric consistency such as AIRS/MODIS or CrIS/VIIRS onboard LEO sensors [14], [15], [16], [17], [18], [19]. And BCOCC has successfully expanded the imager-based OCC approach to GEO sensors.

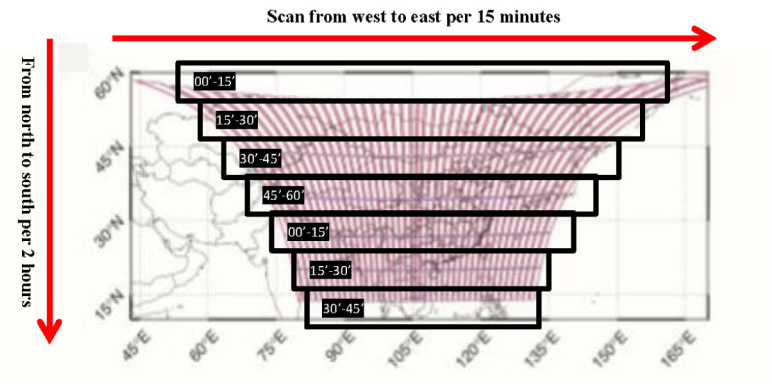
The successfully retrieved CCRs partially fill in the gaps between the clear skies and overcast skies, including overcast cloud edge and broken clouds. CCRs over those clouds will provide useful information for assimilation. Assimilating the LEO CCRs as CLRs has shown positive impacts on improving high impact weather forecasts [17], [18]. Furthermore, appropriately handling the inflated observation errors for LEO CCRs has shown a further positive impact on hurricane forecasts [19]. However, the impacts of assimilating the GIIRS CCRs into NWP models need further investigation.

B. Neighboring Comparisons

The direct evaluation of CCRs is challenging because there are no true observations for comparisons. Two types of indirect evaluations are carried out to understand the quality of the CCRs. Fig. 8 shows the STD of neighboring footprint differences. A pair of neighboring footprints must be the same type of cloud/clear condition for these statistics. For example, both footprints from a pair of neighbors have to be overcast in order for them to be included in the statistics for overcast. For both OCC and BCOCC methods, the low-confidence CCRs (CCflag = 3, green line) overlap with clear sky (red line). The high-confidence CCRs (CCflag = 4, purple line) further improve the STD of neighboring footprint differences, which is much smaller than that of the low-confidence CCRs or clear sky. When calculating the $N*$, the averaged AGRI CLRs collocated with the two adjacent footprints are used [16]. As a

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(a)



(b)

GIIRS observation		AGRI observation timetable (UTC + minutes)											
Number of Regional Scan	UTC	00-05	05-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
i.	0000	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
	0100	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
ii.	0200	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
	0300	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
iii.	0400	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
	0500	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
iv.	0600	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
	0700	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
v.	0800	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
	0900	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
vi.	1000	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
	1100	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
vii.	1200	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
	1300	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
viii.	1400	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
	1500	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
No observation	1600	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
	1700	DISK			No observation			REGC	REGC	REGC	DISK		
	1800	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
ix.	1900	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
x.	2000	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		
	2100	DISK			DISK			REGC	REGC	REGC	REGC	REGC	REGC
xi.	2200	DISK			REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC	REGC
	2300	DISK			REGC	REGC	REGC	REGC	REGC	REGC	DISK		

Fig. 10. (a) FY-4A GIIRS normal observation mode and (b) FY-4A GIIRS and AGRI observation timetable.

result, the gradient of the radiance difference for CCRs should be less or comparable to the clear footprints. Without the QC to ensure the small differences between CCRs and the AGRI CLRs, the CCRs with CCflag = 3 have gradients slightly larger than those with CCflag = 4 of high confidence, but still comparable to clear sky.

Moreover, compared to those from OCC, the CCRs from BCOCC have slightly smaller STDs for the 700–720 and 740–1010 cm^{-1} portions of the spectrum, while slightly larger STDs for the 720–740 and 1010–1130 cm^{-1} portions of the spectrum. The spikes of STDs around 730 and 1100 cm^{-1} are due to the low quality of observed GIIRS radiances in these spectral regions [5], [28]. The 60 GIIRS sounding channels selected for operational data assimilation in CMA's GRAPES are highlighted with gray dots at the bottom in Fig. 8. Low-quality sounding channels are excluded from operational data

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assimilation or retrievals. These results indicate that: 1) a QC of TBRMS less than 1 K is effective in identifying successful CCRs and 2) a QC of absolute BTDs less than thresholds for the three AGRI IR bands is effective in separating high-quality CCRs with CCflag = 4 from low-quality ones with CCflag = 3.

While indirect, these results suggest that the GIIRS CCRs achieve reasonable accuracy in the LWIR spectral region. BCOCC exceeds OCC by generating more accurate CCRs with slightly smaller STDs from neighboring footprints for sounding channels from 700 to 720 cm^{-1} and 740 to 1010 cm^{-1}. Furthermore, by applying the radiance BC to partially cloudy and overcast footprints, more GIIRS observation outliers are flagged by the additional QC of the B12 bias difference. As a result, improvements in STDs for neighboring footprint differences for partially cloudy (blue line) and overcast skies (black line) are evident in Fig. 8.

C. Inter-Comparisons Between GIIRS CCRs and AGRI CLRs

The other indirect way to evaluate CCRs is through the inter-comparison between the convolved radiances of CCRs and the bias-corrected AGRI CLRs. Fig. 9 shows the mean biases and STDs of (AGRI–GIIRS) for CCRs from BCOCC and the sample size (million) according to (a) scene temperature, (b) VZA, (c) latitude, and (d)–(f) FOV number for AGRI bands B12, B13, and B14, respectively. The radiance bias and STD are only converted to BT for visualization.

From Fig. 9(a), the scene temperature has a subtle impact on the mean biases between GIIRS CCRs and AGRI CLRs. The mean biases for CCRs are 0.09, −0.06, and 0.06 K for the AGRI bands B12, B13, and B14, respectively. For scene temperatures between 260 and 320 K, the mean biases for CCRs are within ±0.15 K, for all three AGRI IR bands. The warmer the scene temperature, the larger the mean biases for the AGRI window IR bands B12 and B13, indicating a subtle scene temperature dependence for the mean biases for CCRs. Larger absolute mean biases are found for warmer (>320 K) or colder (<260 K) scene temperatures. For all scene temperatures, the absolute mean biases are within 0.25 K for the window IR bands B12 and B13, and 0.4 K for the CO_2 absorption band B14 in Fig. 9(a). From Fig. 9(b) and (c), the VZA and latitude have no obvious impacts on the mean biases or STDs for CCRs. The absolute mean biases for CCRs are smaller than 0.1 K for all VZAs, and smaller than 0.15 K for all latitudes. Nevertheless, BCOCC ensures the radiometric consistency between the GIIRS CCRs and AGRI CLRs of the three AGRI IR bands, including the AGRI CO_2 absorption band B14.

For the AGRI window IR bands B12 and B13, the absolute mean biases are within 0.2 K for all FOVs, for all CCRs (black line) in Fig. 9(d) and (e). In addition, neither the scene temperature nor the FOV have an obvious impact on the mean biases for CCRs. From Fig. 9(f), for the AGRI CO_2 absorption band B14, the absolute mean biases are within 0.3 K for all FOVs, for all CCRs (black line). A few spikes of mean biases are found for scene temperatures warmer than 300 K. Also, the bias oscillation between the odd and even FOVs is more evident for the AGRI CO_2 absorption band B14 than the two window IR bands B12, and B13. Based on the personal communication with sensor scientists from NSMC, the regular and systematic fluctuation may be due to the sensor issue. After applying the radiance BC, the odd/even fluctuation is largely reduced. Since the observation error is inflated during the CC process [19], the odd/even oscillation for the CO2 absorption band is also evident for CCRs. The exact cause for this kind of odd/even oscillation is still unknown.

For all scene temperatures and all FOVs, the mean biases for all successfully retrieved CCRs from BCOCC, including the high-confidence CCRs (CCflag = 4) and the low-confidence CCRs (CCflag = 3), are less than the NEdTs at mean scene temperature for clear sky, which are 0.2, 0.2, and 0.7 K for the AGRI IR band B12, B13, and B14, respectively. Again, these results indirectly indicate that BCOCC ensures the radiometric consistency between the GIIRS CCRs and AGRI CLRs.

VI. Conclusion

Focusing on the hyperspectral IR sounder CCRs from a GEO platform, this research aims to investigate the feasibility of adapting imager-based OCC approaches from LEO to GEO platforms, more specifically, using AGRI to help cloud clear GIIRS partially cloud radiances. There are two challenges when applying the imager-based OCC approach to GIIRS/AGRI onboard a GEO platform. First, compared with LEO sounders, GIIRS has large gaps between column detectors and nearby detectors within the same column detector. These large gaps may weaken the assumption that the pair of GIIRS footprints have the same radiances in both clear and cloudy portions of the two footprints. Second, the radiometric consistency between GIIRS and AGRI is not as good as those sounders and imagers from LEO platforms, such as CrIS/VIIRS or AIRS/MODIS. From the inter-comparisons between GIIRS and AGRI, the mean biases are dependent on the scene temperatures. For AGRI window IR bands B12 and B13, the colder (or warmer) the scene temperature, the larger the mean negative (or positive) biases. As for AGRI CO_2 absorption band B14, the mean biases are 2–4 K warmer than those of the other two window IR bands. In addition, the mean biases are also dependent on the GIIRS FOVs. The AGRI is slightly colder than the GIIRS for the FOVs near the center, and slightly warmer than the GIIRS for the FOVs near the two ends of each detector column. This is likely due to reduced calibration accuracies for detectors away from the center of the GIIRS sensor array plane. Especially, a bias oscillation between the odd and even FOVs is evident for AGRI CO_2 absorption band B14.

To generate accurate CCRs, the radiometric inconsistency between AGRI and GIIRS must be removed. A radiance BC scheme based on scene temperature and FOV is developed to reduce the relative biases between GIIRS and AGRI. After applying the radiance BC, the scene temperature has a very subtle impact on the residual biases of (AGRI–GIIRS). For scene temperatures between 260 and 320 K, the absolute residual biases are less than 0.03 K for all three AGRI IR bands. Furthermore, the odd/even bias oscillation patterns are largely reduced for AGRI CO_2 absorption band B14. As a result, with the radiance BC for all the clear sky samples,

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the residual biases of (AGRI–GIIRS) are 0.002, 0.003, and 0.015 K, for AGRI bands B12, B13, and B14, respectively. Therefore, the radiance BC method ensures the radiometric consistency between the GIIRS and AGRI for the three AGRI IR bands used, including the CO_2 absorption band B14.

With the help of the bias-corrected AGRI CLRs, the BCOCC method is applied to generate the GIIRS CCRs. The CCRs from BCOCC significantly improve the data yields that can be potentially assimilated as clear sky to the NWP models by 37%, which is three times the CCRs from OCC. Together with the clear sky footprints, around 23% of all GIIRS observations are available for data assimilation into NWP models without worrying about the impact of clouds. From the STDs of neighboring footprint differences, the GIIRS CCRs achieve accuracy comparable to that from clear skies in the LWIR spectral region. BCOCC exceeds OCC by generating more accurate CCRs for sounding channels from 700 to 720 cm^{-1} and 740 to 1010 cm^{-1}, with slightly smaller STDs in neighboring footprint differences. In addition, the convolved radiances of GIIRS CCRs are inter-compared to the bias-corrected AGRI CLRs to understand the quality of the CCRs. The mean biases for CCRs are 0.09, −0.06, and 0.06 K for the AGRI bands B12, B13, and B14, respectively. Nevertheless, BCOCC ensures the radiometric consistency of the GIIRS CCRs and AGRI CLRs, for the three AGRI IR bands. By appropriately handling the inflated observation errors, the CCRs can be effectively assimilated into NWP models [19]. To quantify the impacts of large time differences will be an interesting topic for future work. In addition, the impacts of assimilating the GIIRS CCRs into NWP models need further investigation.

To make AGRI and GIIRS radiances consistent, the GSICS procedure can be adopted by inter-calibrating these two sensors with the same standard reference, such as the infrared atmospheric sounding interferometer (IASI). This will also avoid the need to estimate scene-dependent BC. However, due to diurnal variations of the radiance biases between AGRI and GIIRS [27], the BC procedure based on inter-comparisons might still be needed. The GSICS technique using a high-quality hyperspectral IR sounder onboard LEO satellite only ensures consistent radiometric accuracy at the local passing time, i.e., 9:30 AM or 9:30 PM for IASI. In other words, the radiometric inconsistency between AGRI and GIIRS at other times may not be eliminated completely from the GSICS technique. Furthermore, study have shown that the operational FY-4B/GIIRS has a better performance compared to the experimental FY-4A/GIIRS, and the calibration performance of FY-4B/GIIRS demonstrates a reduced scene BT dependency [35]. A similar approach of CCR generations could be adopted to the future FY-4 satellites.

This study demonstrates the importance of placing an advanced IR sounder and imager onboard the same GEO platform for better applications. The methodology developed for the GIIRS/AGRI CCRs is highly relevant to future GEO hyperspectral IR sounder missions. There are three missions in the near future, including Europe's MTG [36], [37], Japan's Himawari-10 [38], and the United States' GeoXO [32]. They will carry a hyperspectral IR sounder with a spatial resolution and ground sample distance of around 4 km. The much smaller ground sample distance will better meet the assumption of the same thermodynamics in the clear or cloudy portion of two nearby footprints. This will be particularly useful for the Himawari-10 mission where both the sounder and imager are on the same platform [38]. For MTG [36], [37], the two sensors will be on different platforms, but three of the MTG imagers will have the same longitude of 0° as the sounder. So, the CCRs can be generated with careful collocation between the sounder and imager. For GeoXO [32], the sounder satellite will have a longitude of more than 30° away from the imager satellites. They will observe the same atmosphere and clouds from different angles. The angle discrepancy between the two sensors will violate the assumption of the same thermodynamics in both the clear and cloudy portions of two nearby footprints. Additionally, there will be limited regions where the imager can be used to characterize the CLRs within a sounder footprint due to the footprint size increasing with zenith angle. These difficulties may limit the application of the CC method for the GeoXO sounder.

APPENDIX

A. FY-4A Observation Timetable

Fig. 10(a) shows FY-4A GIIRS normal observation mode and (b) FY-4A GIIRS and AGRI observation timetable.

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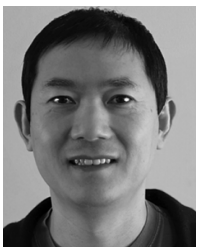
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